CLINICAL TRIAL FAILURES · MISSED ENDPOINTS · PSYCHIATRY & MENTAL HEALTH PHASE 2,3 · STARTS 2015–2024

Serotonin 5-HT2A: 10 of 21 completed trials missed their primary endpoint

10 completed trials across 4 sponsors and 7 molecules ran to the end and missed their primary endpoint on the sponsor's own posted result; 9 met it. Only 2 of 87 closed trials were stopped early for efficacy or safety, which is why this class is read through its completed trials rather than a stop rate. Most completed trials post nothing readable, so the count is a floor.

10 of 21
completed trials with a readable primary result missed it · 9 met it

4 / 7
sponsors / molecules behind the misses

2.3%
stopped early: 2 of 87 closed trials (95% CI 0.6%–8.0%)

71.3%
of 122 trials have closed · 35 still open or unresolved

10 of 21 **Completed and missed the primary endpoint.** Of 74 completed trials in this segment, 21 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 10 missed and 9 met, 2 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 16 of the 68 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

The completed trials that missed

Trial	Ph	Sponsor	Experimental drugs	Started	What the posted result says
NCT05523895	3	ACADIA Pharmaceuticals Inc.	PIMAVANSERIN	2022-08	Change From Baseline at Week 6 in Caregiver-rated Aberrant Behavior Checklist Irritability (ABC-I) Subscale Score — p 0.2986, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Mean Difference (Final Values) -1.6 (95% CI -4.6 to 1.4). Method: Mixed Models Analysis.
NCT04531982	3	ACADIA Pharmaceuticals Inc.	PIMAVANSERIN	2020-08	Negative Symptom Assessment-16 (NSA-16) Total Score - Change From Baseline to Week 26 — p 0.4825, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Method: Mixed Models Analysis.
NCT04174365	3	Otsuka Pharmaceutical Development & Commercialization, Inc.	BREXPIPRAZOLE	2019-10	Mean Change From Baseline to Week 8 in Aberrant Behavior Checklist - Irritability (ABC-I) Subscale Score — p 0.4597 against the sponsor's stated threshold of 0.05: not significant. Least squares (LS) mean difference -1.22 (95% CI -4.49 to 2.05). Method: Mixed model repeated measures(MMRM).
NCT03968159	3	ACADIA Pharmaceuticals Inc.	PIMAVANSERIN	2019-04	Change From Baseline to Week 5 in Hamilton Depression Scale (17 Items) (HAMD-17) Total Score — p 0.2956, no threshold stated; 0.1 or above, so not significant under any conventional threshold. LSM difference -0.9 (95% CI -2.5 to 0.8). Method: Mixed-effects model for repeated measure.
NCT03538691	3	Otsuka Pharmaceutical Development & Commercialization, Inc.	BREXPIPRAZOLE, Antidepressant therapy	2018-07	Phase C: Time-to-Relapse by Any Criteria as Defined in Blinded Addendum — p 0.51, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Hazard Ratio (HR) 1.138 (95% CI 0.776 to 1.669). Method: Log Rank.
NCT03257865	3	Otsuka Pharmaceutical Development & Commercialization, Inc.	BREXPIPRAZOLE	2017-09	Change From Baseline In Young-Mania Rating Scale (YMRS) Score At Week 3 — p 0.1011, no threshold stated; 0.1 or above, so not significant under any conventional threshold. Treatment difference -1.62 (95% CI -3.56 to 0.32). Method: mixed-effect model repeated measure.

The 6 most recently started of 10 completed trials that missed. The diligence report lists every one, with the numbers or the sentence each verdict was read from.

How this compares

Serotonin 5-HT2A



2.3%

2/87 closed

all psychiatry & mental health trials with a resolved drug target



3.7%

15/409 closed

All psychiatry & mental health Phase 2,3



2.9%

30/1022 closed

What is in this class: 110 distinct experimental drugs across 122 trials (small molecule 122, unknown 10). Most tested: PSILOCYBIN (35), LUMATEPERONE (17), BREXPIPRAZOLE (17), PIMAVANSERIN (14), RISPERIDONE (12). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 2 stops came from 2 sponsor-asset programmes across 2 sponsors; the largest (Supernus Pharmaceuticals, Inc.) contributed 1; removing that programme's trials from both sides leaves 1.2%. Screened alongside 12 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 4 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 6.9%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	2	34	5.9%	1.6%–19.1%
2018–2020	0	28	0.0%	0.0%–12.1%
2021–2024	0	25	0.0%	0.0%–13.3%

Where the stops sit

Efficacy / futility: 2.
Sponsors with most stops: Supernus Pharmaceuticals, Inc. (1), H. Lundbeck A/S (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02691182	3	Supernus Pharmaceuticals, Inc. SUPN	MOLINDONE	2021-05	Efficacy	The program was shut down due to a lack of efficacy.
NCT03230864	3	H. Lundbeck A/S	Lu AF35700, RISPERIDONE, OLANZAPINE	2018-12	Efficacy	New data; the study was terminated based on new efficacy data from another study

Method
Denominator: ClinicalTrials.gov interventional Phase 2,3 psychiatry & mental health trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 44% of industry psychiatry & mental health trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits
Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 71.3% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.